

Arduino UNO Q Linux OS Master Guide

Zero-to-Hero Survival Guide for Qualcomm Dragonwing Debian Linux (In Hinglish & English)

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1. Arduino UNO Q Mein Linux Kaise Kaam Karta Hai?

Arduino UNO Q koi ordinary board nahi hai — yeh ek Dual-Brain Supercomputer board hai:

- Brain 1 (STM32U585 MCU): Yeh normal Arduino jaisa hai jo C++ sketches chalata hai aur sensors/motors ko microsecond precision se control karta hai.
- Brain 2 (Qualcomm Dragonwing QRB2210 MPU): Yeh poora ek Quad-Core 64-bit Computer hai jisme Debian Linux OS chalta hai. Isme Camera (MIPI-CSI), GPU AI Neural Models, Audio, aur Docker chalta hai!

Aapko darne ki bilkul zaroorat nahi hai — Linux par kaam karna bilkul WhatsApp ya browser use karne jaisa simple hai agar aapko 5-6 basic commands pata hon!

2. Arduino UNO Q Ke Linux Terminal Mein Login Kaise Karein?

Board ke Linux OS ko control karne ke 2 sabse aasan tareeqe hain:

Tareeqa A: Wi-Fi / Ethernet SSH se (Sabse Best)

1. Apne Laptop/PC par Command Prompt (CMD) ya PowerShell open karein.
2. Yeh command type karke Enter dabayein:

```
$ ssh arduino@arduino-uno-q.local
```

🔗 Matlab: Board ke Linux mein login karne ke liye.

3. Password maangega toh type karein: arduino (Password screen par dikhega nahi, bas type karke Enter dabana).

Tareeqa B: Arduino App Lab / VS Code Terminal se

Arduino App Lab software open karein, apna board select karein aur 'Open Linux Terminal' button par 1-click karein — terminal automatically khul jayega!

3. Top 10 Daily-Life Linux Commands (Cheat Sheet)

Command	Kaam (Action)	Aasan Hindi Matlab / Example
ls	List files	Dekho iss folder ke andar kaun-kaun si files hain.
ls -la	Detailed List	Files ka size, permissions aur hidden files bhi dekho.
cd folder_naam	Change Directory	Kisi folder ke andar ghuso (jaise 'cd NeuroSign_HMI').
cd ..	Go Back	Ek step peeche/parent folder mein wapas aao.
pwd	Print Working Dir	Main abhi computer ke kis folder mein

sudo command	Superuser Do	khada hoon woh batao. Administrator / Boss permission ke saath command chalao.
mkdir naya_folder	Make Directory	Naya folder banao.
nano file.txt	Text Editor	Terminal ke andar hi kisi file ko padho ya edit karo.
htop / top	Task Manager	Dekho CPU kitna use ho raha hai aur RAM kitni bachi hai.
exit	Log Out	Linux terminal session band karke wapas aa jao.

4. NeuroSign-HMI Project Run Karne Ke 4 Golden Commands

Jab aap board par login kar lein, toh sirf yeh 4 steps run karne hain:

Step 1: Project Code Download / Clone Karna

```
$ git clone https://github.com/voidreformer/NeuroSign-HMI.git
```

🔗 Matlab: GitHub se poora project board par download ho jayega.

```
$ cd NeuroSign-HMI/docker
```

🔗 Matlab: Docker environment folder ke andar chale jao.

Step 2: Docker AI Container Start Karna

```
$ docker compose up -d --build
```

🔗 Matlab: OpenCV, MediaPipe, Adreno GPU Delegates aur Audio engine background mein start ho jayenge.

Step 3: Master Orchestrator (Main App) Launch Karna

```
$ docker exec -it neurosign_engine python3 /app/app_mpu/main_orchestrator.py
```

🔗 Matlab: Camera on ho jayega, 15 gestures AI classification aur speaker active ho jayega!

Step 4: Script Ko Stop / Close Karna

Jab testing poori ho jaye aur script band karni ho, toh keyboard par bas yeh dabayein:

```
$ Ctrl + C
```

🔗 Matlab: Running Python script turant safely terminate ho jayegi.

5. Common Mistakes & Troubleshooting (Quick Fixes)

Q1: 'Permission denied' error aa raha hai?

🔑 Fix: Command ke aage 'sudo' laga do (jaise: `sudo docker compose up -d`).

Q2: Camera feed nahi dikh raha ya /dev/video0 error?

🔑 Fix: Check karo camera ribbon cable sahi se lagi hai ya nahi. Terminal par check karein:

```
$ ls /dev/video*
```

🔗 Matlab: Dekho /dev/video0 list ho raha hai ya nahi.

Q3: Speaker se aawaz nahi aa rahi?

👉 Fix: Check karo MAX98357A I2S amplifier ke wires sahi pin par hain. Test sound play karein:

```
$ aplay /app/audio_prompts/english/01_emergency_sos.wav
```

🔗 Matlab: Direct speaker par test voice play hogi.

Q4: Terminal par screen freeze ho gayi ya script atak gayi?

👉 Fix: Keyboard par `Ctrl + C` dabayein ya doosri window se `docker restart neurosign\_engine` run karein.